

[DC] Fostering Well-being with Virtual Reality Applications

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ABSTRACT

My research focuses on how mental health and well-being can be supported by Virtual Reality (VR) interventions. In particular, as a first step in my research, I identified three main research gaps. In my PhD, I design, develop and evaluate VR prototypes, and derive specific design recommendations for VR interventions that aim to improve well-being. In particular I explore how VR can effectively i) provide autonomy to users, e.g. for emotional expression, ii) can combine cognitive and affective approaches to foster understanding of health as a holistic concept, e.g. through metaphoric visualisation of health data, and iii) provide guidance when reflecting and practising well-being strategies.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI—

1 INTRODUCTION AND MOTIVATION

Mental health and well-being support are commonly addressed design goals in Human-Computer Interaction (HCI) research (e.g. [2, 16]). Mental health is an integral part of health and forms the foundation for well-being [23], which is an abstract construct that is defined as *a dynamic optimal state of psychosocial functioning* [3]. In contrast to existing technologies that improve well-being, such as stress trackers, desktop or mobile interventions, applications in virtual reality (VR) allow for deep immersion in a virtual environment (VE). VR can reproduce the complexity of real-life situations while variables such as sound, light, and movement of objects can be close to completely controlled to provide meaningful interventions.

The way that all those technologies, and especially VR, support mental health and well-being is as manifold as the area is broad, encompassing therapeutic interventions to treat anxiety (e.g. [10]), mindfulness training (e.g. [12]), stress reduction support (e.g. [22]), VR as positive technology (e.g. [8]) and for transformative change [5], VEs for different forms of journaling (e.g. [19]), reflection support [1] and emotion regulation interventions [11], among others. In contrast to those works, which have endeavoured to understand the benefits and design space of VR in each particular area, I aim to consolidate knowledge, exploring the design space of VR applications for mental health and well-being from a broader and more holistic perspective. My motivation stems from mismatches between what psychotherapists envision VR applications for mental well-being to offer and what commercial applications actually provide [20]. I identified three major research gaps, which I will address with my PhD. They are motivated by related work and own research [20], which will be presented in short below.

- i) designing for autonomy and individual expression, e.g. through autonomous emotional expression. Users' autonomy is mostly limited to act within a given virtual environment (VE), e.g. by manipulating objects or choosing between different preset VEs. In the few cases where users are able to create

their own VE (e.g. [4]) beforehand, the process is cumbersome for designers and users remain passive during the actual experience. However, autonomy as a basic psychological need underlies personal growth [15], which ties in with well-being, and should thus be considered more in VR experiences.

- ii) designing for mental health and well-being as a holistic concept, e.g. synergy of body and mind. It is challenging for a lot of people to identify what they are feeling and what their physical reactions mean in regard to their health [2, 6]. To mitigate, smartwatches can provide an overview of that health data useful as a first step for self-care. However, research in personal informatics indicates that current standard representations for data, mainly 2D graphs, are unlikely to effectively foster engagement and reflection [1, 6, 13]. VR may mitigate this challenge by presenting an effective and immersive approach to represent data. Yet, the few approaches that involve VR (e.g. Yoo et al. [24]) still provide quantitative feedback in form of graph and number representation although the range of visualising data in VR is larger and can be more abstract compared to desktop or mobile systems. Unique affordances of VR should be considered more in the design process of VR experiences in order to design much needed fruitful interventions [16].
- iii) designing for cognitive support, e.g. reflection. Reflection - loosely defined as understanding [17] - benefits health, well-being and personal growth (e.g. [18]). Yet, users need to be actively encouraged and guided [18]. Although there has been work on providing a conceptual model to understand how VR applications should support reflection [7], it does not provide recommendations for designing reflective guidance in VR.

Outcomes of the research at hand will deepen our understanding of the lived experiences when using VR for well-being support. Findings of my dissertation will be important for the design of future VR applications in the field of identifying and expressing emotions, understanding personal informatics such as health data, and supporting reflection.

2 RESEARCH QUESTION

This thesis seeks to define the design space of VR applications for mental well-being from a holistic perspective. I pose the following research question, based on three identified research gaps: *How should VR applications for mental well-being be designed to i) empower users and offer autonomy, ii) support users to connect with processes happening in their body and mind in an engaging way, and iii) provide guidance to enhance well-being.*

3 RESEARCH DESIGN

This research presents findings from two empirical studies, each seeking to provide different, yet complimentary, design recommendations.

As a first step, an extensive expert interview with psychotherapists was conducted with the aim to explore which features and objectives mobile and VR applications should offer to provide good well-being support. Based on those findings, I conducted a systematic review of consumer applications in order to identify (mis-)matches in 16

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VR applications. I found that existing VR applications should better provide individually fitting approaches, combine aspects of the body and the mind, and guide users towards a meaningful reflection [20].

Based on those research gaps, I build a VR application called 'Mood Worlds', modified from 'Open Brush', in which offering autonomy was the specific focus. Users created their own VE through a broad variety of brushes, environments, and manipulable objects. In a user study (N=16) Mood Worlds was evaluated for benefits for emotional expression. Participants were positively affected by this experience, and especially enjoyed the spatiality and movement possibilities within VR [21].

4 EXPECTED NEXT STEPS

In a next step and using Mood Worlds as basis, the element of providing guidance in VR will be explored. Guidance provides scaffolds to thought processes [14], supports an effective reflection [18], enhancing well-being. First, I want to explore qualitative visualisations. I will analyse methods used in psychotherapy and translate (and enhance) them to VR. Through those, users will be guided visually in the process of accepting negative emotions. Further, it will be investigated how weather can be used to help users reflect on their health data such as stress levels. In contrast to static 2D representations of stress, being immersed in a qualitative representation of those data might increase reflection and engagement with one's physical and mental health. Second, I will analyse the effects of voice-based guidance in VR. In a non-VR setting, conversational interfaces, chatbots or voice-assistants were found to provide valuable scaffolds to thoughts (e.g. [14]) and for facilitating reflection (e.g. [9]). The design of such voice-based guidance will be informed so that individuality and autonomy is placed into focus.

5 EXPECTED BENEFITS FROM PARTICIPATION IN IEEE VR'23 DOCTORAL CONSORTIUM

I started my PhD during the pandemic, and although I am honoured to be supervised by two professors (Johannes Schöning, HSG Switzerland and Yvonne Rogers, UCL London), I rarely had the possibility to get input from the HCI community on my topic. The IEEE VR Doctoral Consortium will take place during the third year of my PhD, and as such it presents itself as a great opportunity to discuss the challenges of understanding my empirical findings in a broader perspective. I believe that the consortium will provide constructive feedback and inspiration on how to best refine and present the arguments of the thesis as they cover a broad range of sub-topics. This will help to form a cohesive and strong line of argumentation.

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